PHY · Outside the 2026–2027 syllabus — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Standard Model and other content no longer in the syllabus

Syllabus: not in the current syllabus

Frequency: 10 of 10 exam years; 115 marks in all.

Marks reward: Recall 49, Final answer 20, Qualitative explanation 19, Substitution and working 17, Select the principle or relationship 4, Evaluate and conclude 3, Diagram 3

Past questions: 2016C-21(b); 2016P-17(ai); 2016P-17(aii); 2016P-17(b); 2017C-21(e); 2017P-15(ai); 2017P-15(bi); 2017P-15(bii); 2017P-16(a); 2017P-16(bi); 2017P-16(d); 2018C-18(b); 2018P-17(a); 2018P-17(b); 2018P-17(c); 2018P-17(d); 2019C-20(cii); 2019P-15(a); 2019P-15(b); 2019P-15(c); 2020C-21(b); 2020P-14(a); 2020P-14(d); 2020P-19(d)…

Summary: This point covers Standard Model particle physics and a few astrophysics items that have been removed from the current Year 12 syllabus, such as quark composition, particle classification, conservation laws, force carriers, and galaxy rotation or dark matter. Past papers used it mainly for short recall, identification and one-line explanation questions, so it is worth recognising as legacy content but not expecting on current papers.

Recognise: Questions that ask for the quark makeup of a named hadron or antiparticle, classify a particle as a baryon, meson, lepton or boson, check conservation of baryon number, lepton number or charge in a decay equation, name the mediator of a force, or explain a galaxy's rotation curve or Hubble-type velocity observations.

Steps

1. Identify the particle from its quark content or its listed properties, and remember that an antiparticle has antiquarks with reversed charge (for example, a proton is uud and a neutron is udd).
2. For a conservation check, total each quantity on each side of the equation: baryon number (quarks +1/3, antiquarks -1/3), lepton number (leptons +1, antileptons -1, all other particles 0), and electric charge.
3. Compare the two totals. If they match, the quantity is conserved; if they differ, identify the missing particle needed to balance the equation.
4. Link each force to its mediator: strong force to the gluon, weak force to the W or Z boson, electromagnetic force to the photon, and the proposed gravitational mediator to the graviton.
5. When asked to justify a property of a force carrier, connect it to the force's range and behaviour: an unlimited range implies a massless mediator, and a force that never repels equal charges implies no odd spin.
6. For galaxy or Hubble questions, state the observation first (for example, velocities on opposite sides of a rotating galaxy), then the inference (for example, unseen dark matter or the direction of motion from Doppler shift).

Where the marks are

- A mark for each correct quark composition, particle symbol, charge or classification, which is mostly recall.
- A mark for each side's total in a conservation calculation, and a separate mark for the conclusion about whether the quantity is conserved.
- A mark for naming the correct mediator for a force, with a second mark for a valid reason where a property is asked for.
- A mark for stating the observation and the inference together in astrophysics explanations, not the inference alone.

- A single-word answer is often enough when the question asks only to identify a particle or force.

Pitfalls

- Dropping the bar on an antiquark or assigning the wrong sign to a quark's charge or baryon number.
- Calling a kaon a baryon. Mesons contain one quark and one antiquark; baryons contain three quarks.
- Checking only one side of a conservation equation, or forgetting that the neutrino in a decay has a lepton number.
- Giving a mediator's mass or spin without a reason, which loses the explanation mark.
- Presenting the graviton as an observed particle when it is a proposed, undetected mediator.